Sanyam Jain

PhD Fellow – Generative AI for Dental and Medical Imaging
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RESEARCH PROFILE

PhD fellow developing generative AI for dental and medical imaging: generative adversarial networks, denoising diffusion probabilistic models, super-resolution and segmentation-guided synthesis, applied to 2D panoramic radiographs and 3D dental point clouds. Research addresses data scarcity and class imbalance in clinical imaging by synthesising anatomically consistent images and using them as augmentation for downstream classification, detection and segmentation. Additional work on federated, centralised and local learning, continual learning and machine unlearning. **20+ publications; 155 citations; h-index 7; i10-index 5** (Google Scholar, 27 July 2026).

EDUCATION

2024 – 2027 (expected)	**PhD, Health Science** – Department of Dentistry and Oral Health, Aarhus University, Denmark. Thesis: *Synthetic Dental Imaging using Generative Artificial Intelligence*. Supervisor: Assoc. Prof. Ruben Pauwels. Co-supervisors: Postdoc Bruna Neves de Freitas (AIAS-AUFF Fellow, Aarhus University); Prof. Alexandros Iosifidis (Tampere University, Finland). Mid-term evaluation completed March 2026; 25.7/30 ECTS of PhD coursework completed.
2022 – 2024	**MSc, Applied Computer Science** – Østfold University College (HiØ), Halden, Norway.
2020 – 2022	**Direct PhD programme (discontinued), Quantum Information and Computation** – Indian Institute of Technology Jodhpur, India.
2015 – 2019	**BTech, Computer Science and Engineering** – University of Petroleum and Energy Studies, Dehradun, India.
2024 –	**Danish language** – Danskuddannelse 3, A2B Sprogskole Århus; DU3.2 completed January 2026, DU3.3 ongoing.

RESEARCH POSITIONS

Sep 2024 – present	**PhD Fellow**, Department of Dentistry and Oral Health, Aarhus University, Denmark. Generative AI for dental radiography (Pano-GAN, PanoDiff-SR, CarDiff, third-molar synthesis); design, training and evaluation of large generative models and downstream diagnostic models.
Feb – Apr 2026	**Visiting PhD Researcher**, Tampere University, Finland (Prof. Alexandros Iosifidis). Model robustness for lifelong learning; continual learning and machine unlearning for selective data deletion.
Jul – Aug 2024	**Summer Research Intern**, Norwegian University of Science and Technology (NTNU), Norway. Education Technology Lab – computer vision and large vision models.
2023 – 2024	**Research Assistant**, OsloMet – Oslo Metropolitan University, Norway (Jun–Aug 2023; Sep 2023 – Mar 2024). Complexity in discrete cellular automata; capturing emergent behaviour with evolutionary computation.
2022 – 2024	**Research Assistant (Forskningsassistent)**, Østfold University College, Norway (Oct 2022 – Jun 2023; Mar – Jun 2024). Open-endedness, artificial life and biologically inspired AI.
2021 – 2022	**Teaching Assistant**, IIT Jodhpur – Machine Learning (Winter 2022); Foundations of Quantum Information and Computation (Autumn 2021).
2018 – 2020	**Industry**: Alexa / conversational-AI developer, Keyring Corp (AWS Activate startup); RPA consultant (UiPath, self-employed); data analysis trainee, Fiserv; software engineering intern, YellowAnt; remote intern, IBM Watson.

COMPUTATIONAL AND METHODOLOGICAL COMPETENCIES

- **Generative modelling:** GANs (Wasserstein loss, gradient penalty), denoising diffusion probabilistic and latent diffusion models, conditional and segmentation-guided synthesis, image inpainting, super-resolution, 2D/3D point-cloud diffusion.
- **Downstream tasks and evaluation:** classification, object detection and semantic segmentation (Attention U-Net, ResNet, EfficientDet, YOLO); FID, Inception Score, t-SNE, SSIM/PSNR/LPIPS/DISTS; design and statistical analysis (ROC/AUC) of expert observer studies with clinicians.
- **Distributed and privacy-preserving learning:** federated, centralised and local learning paradigms; continual learning and machine unlearning.
- **Tools and infrastructure:** Python, PyTorch; multi-GPU deep-learning training on Linux compute clusters; large-scale image dataset curation and preprocessing; reproducible pipelines with public release of code and synthetic data.

TEACHING, SUPERVISION AND DISSEMINATION

2026 –	Co-supervisor, MSc project on 3D dental mesh segmentation, University of São Paulo, Brazil; co-supervisor, BSc project on 3D dental point-cloud synthesis, School of Natural Sciences, Aarhus University (with Department of Mathematics).
2025	Hands-on teaching and dissemination sessions for students and researchers (43+ registered hours), including "Basics of Generative AI" for the Intelligent Systems group and the "Virtual Patient" project with Prof. Eduardo.
2025	Oral presentation, ECAI 2025 – HAIC workshop, Bologna, Italy. Regular participant in departmental journal clubs and PhD Day.

INTERNATIONAL COLLABORATION AND ACADEMIC SERVICE

- Contributor, ITU/WHO/WIPO Global Initiative on Artificial Intelligence for Health – Topic Group on Dental Diagnostics and Digital Dentistry (TG-Dental, FG-AI4H).
- Active research collaborations with Tampere University (FI), OsloMet and Østfold University College (NO), ACTA Amsterdam (NL), University of São Paulo (BR), and clinical partners in 9 countries on the multi-centre third-molar synthesis study.

SELECTED PHD TRAINING AND CERTIFICATIONS

- **PhD courses (25.68/30 ECTS):** Computer Vision (10 ECTS); IBM Generative AI Engineering Professional Certificate (7.28 ECTS); Artificial Intelligence for Scientific and Academic Writing (2.5 ECTS); Commercial Research Collaboration (1.6 ECTS); Become a Productive and Joyful Researcher (1.5 ECTS); Project and Time Management (1.0 ECTS); Research Data Management, FAIR Principles and Open Access (0.3 ECTS).
- **Certifications:** NVIDIA DLI – Building Transformer-Based Natural Language Processing Applications; Generative AI Language Modeling with Transformers; Generative AI Foundational Models for NLP and Language Understanding; Data Analysis with Python; UiPath RPA (UiRPA, UiARD).

CURRENT AND PLANNED RESEARCH DIRECTIONS

- **Conditional synthesis of pathology:** segmentation-guided diffusion for dental caries (CarDiff) and for rare third-molar / mandibular-canal anatomical relationships, followed by augmentation studies quantifying the gain in segmentation and classification accuracy.
- **3D dental point clouds and meshes:** low-resolution point-cloud diffusion followed by super-resolution, towards synthetic 3D dental data for mesh segmentation and treatment planning.
- **Human feedback in the synthesis loop:** reinforcement learning from clinician feedback to steer generative models away from anatomically implausible output.
- **Evaluation and interpretability:** trajectory and drift analysis in diffusion models (InJecteD); benchmarking cultural and linguistic bias of large language models in dentistry.

LANGUAGES

English (academic working language) • Danish (Danskuddannelse 3.2 completed January 2026, DU3.3 ongoing) • Hindi.

A list of the 10 most relevant publications is submitted as a separate appendix, as required by section 3.1 of the H1-2027 call.